

S^n , ~~S^1~~ , RP^n , $\langle \dots \rangle$
 $(n \geq 1)$
 $\downarrow$
 $\pi_1 = 0$

2-dim mfd $\underline{L(p,q)}$

$M \cup N$ $\wedge \vee$ $\langle S, S' \rangle / \langle e^{2\pi i p}, e^{2\pi i q} \rangle$
 τ_0, τ_1
 $a \xrightarrow{\tau_0} b$ $a \xrightarrow{\tau_1} b$
 $\tau_0 \sim \tau_1$
 $\mathbb{Z} \times \mathbb{Z}$

$\langle a, b \mid aba = b \rangle$
 $ababab \dots$
 $abab^t = e$
 $ab = ba \rightarrow \text{group } \mathbb{Z} \times \mathbb{Z}$
 $\langle a, b \mid ab = ba \rangle$
 $\mathbb{Z} \times \mathbb{Z}$

$S^1 \vee S^2$
 $T: \mathbb{Z} \times \mathbb{Z}$
 $T \hookrightarrow \mathbb{Z} \times \mathbb{Z}$
 $\mathbb{Z} \hookrightarrow \mathbb{Z} \times \mathbb{Z}$
 $\mathbb{Z} \times \mathbb{Z}$

Theorem
 $M: n\text{-dim}$
 with $(n \geq 2)$
 then $\pi_1(M) \cong \pi_1(M \setminus \{x\})$
 $\pi_1(RP^n) \cong \mathbb{Z}_2$
 $(n \geq 2)$
 $\pi_1(M) \cong \pi_1(M \setminus \{x\})$
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$\langle a, b \mid a^2 = b^2 \rangle$
 $\mathbb{Z} \times \mathbb{Z}$
 $\mathbb{Z} \times \mathbb{Z}$

$S^2/p = S$
 $\mathbb{Z} \times \mathbb{Z}$

Example
 $S^2/p = S$
 $\mathbb{Z} \times \mathbb{Z}$

S^2 : 0-cell $\mathbb{Z}$
 $1\text{-cell } \mathbb{Z}$
 $2\text{-cell } \mathbb{Z}$

RP^2 : 0-cell: $\mathbb{Z}$
 $1\text{-cell: } \mathbb{Z}$
 $2\text{-cell: } \mathbb{Z}$

$RP^2: \mathbb{Z}_2$ $\langle a \mid a^2 = 1 \rangle$
 $a^2 = 1$
 $RP^2: 0\text{-cell: } \mathbb{Z}$
 $1\text{-cell: } \mathbb{Z}$
 $2\text{-cell: } \mathbb{Z}$

$\pi_1(RP^2) = \pi_1(RP^2) = \mathbb{Z}_2$
 $\pi_1(RP^2) = \pi_1(RP^2) = \mathbb{Z}_2$

$\mathbb{R}/\mathbb{Z}$
 $\mathbb{R}/\mathbb{Z}$

$\mathbb{Z} = \mathbb{Z} + \mathbb{Z}$
 $\mathbb{Z} = \mathbb{Z} + \mathbb{Z}$

$\int \frac{1}{x} dx = \ln|x| + C$
 $\int \frac{1}{x} dx = \ln|x| + C$